

Neural Texture Compression

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[Extended Writeup](#)

[Full code and reproduction instructions \(Github\)](#)

P2. S3TC (DXT1) baseline

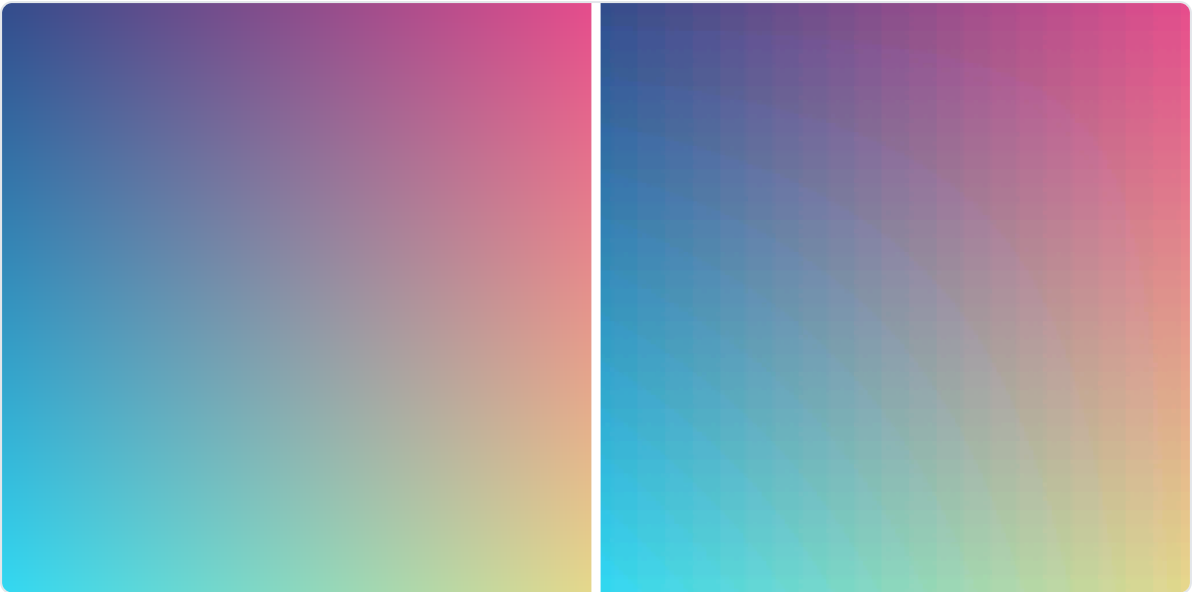
How the endpoints were chosen

For every 4×4 block:

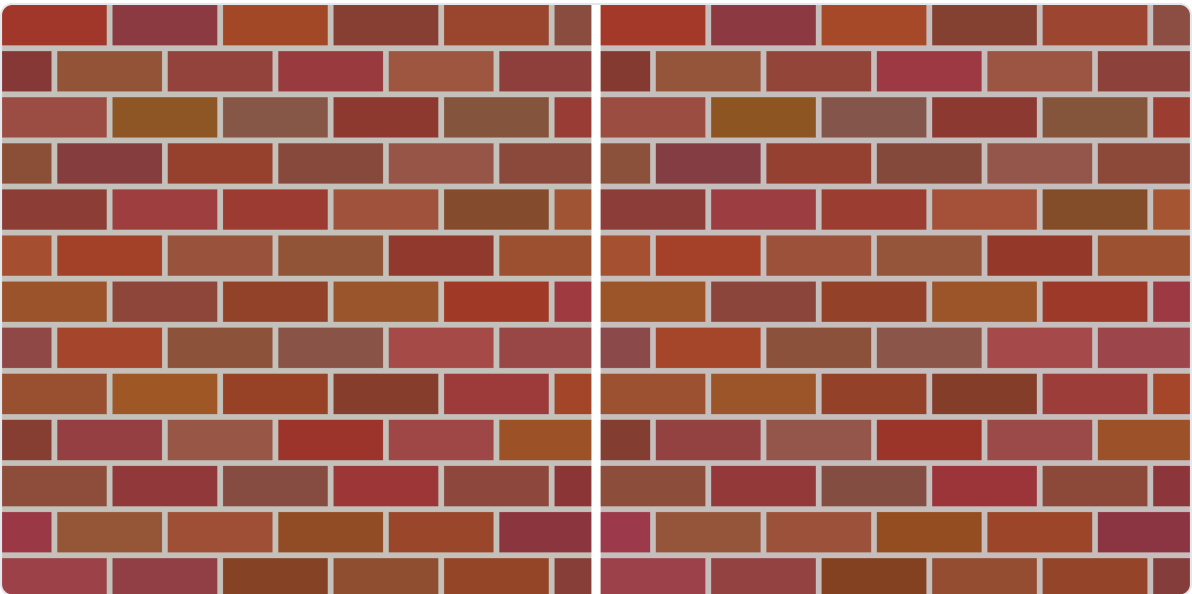
- PCA: subtract the block mean, take the dominant direction from a few power iterations on the 3×3 covariance, project the 16 texels, and use the two extremes as endpoints.
- Refit: assign each texel to its nearest palette entry, then solve the closed-form 2-variable least-squares problem for the endpoints that best reconstruct those colors; repeat assign-and-refit twice.
- Flat blocks keep their previous endpoints (the fit is underdetermined). Endpoints are then quantized to RGB565 and written in 4-color mode.

PSNR and compression

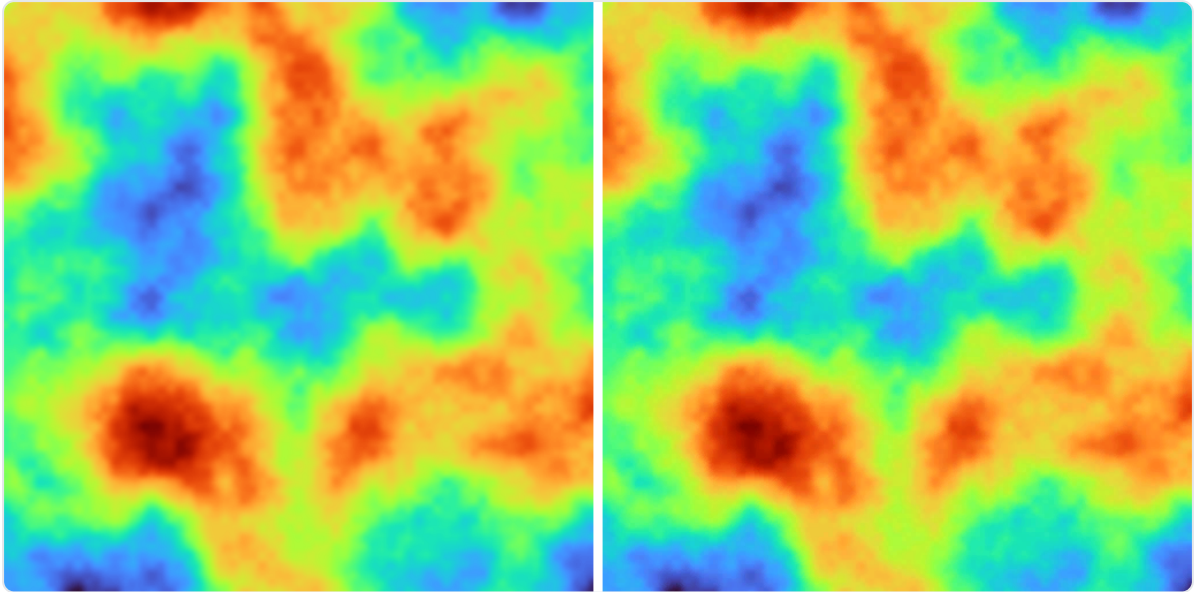
Texture	PSNR (dB)	Compressed	Original	Ratio (comp/orig)
gradient	42.89	128 KB	768 KB	1/6 (0.167)
bricks	41.39	128 KB	768 KB	1/6 (0.167)
clouds	42.63	128 KB	768 KB	1/6 (0.167)



Gradient: original (left) vs S3TC (right).



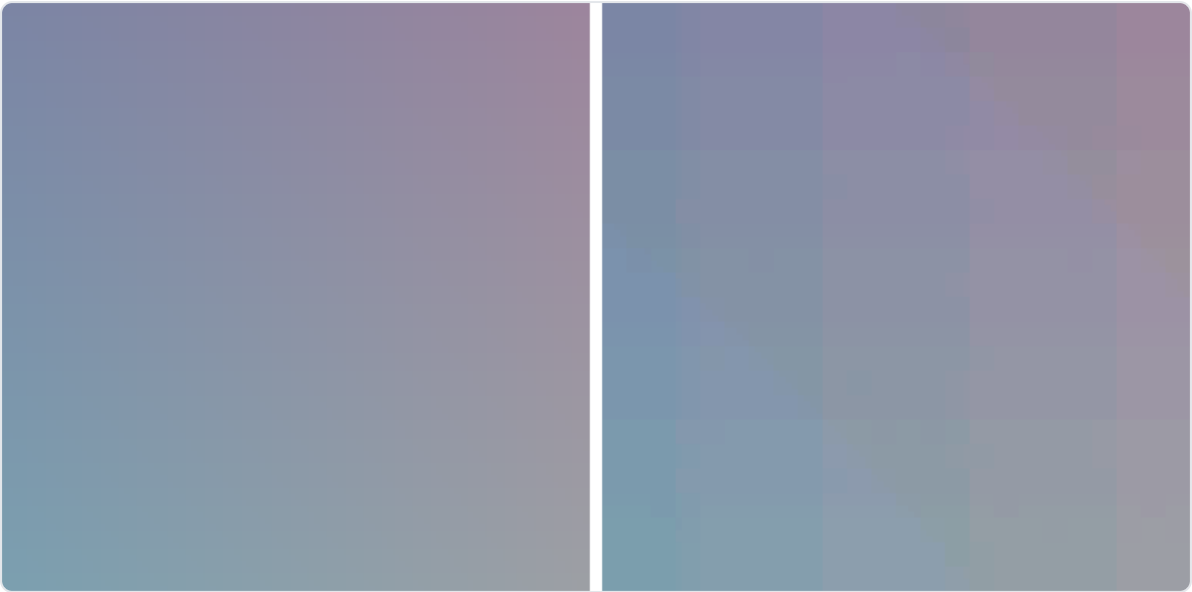
Bricks: original vs S3TC.



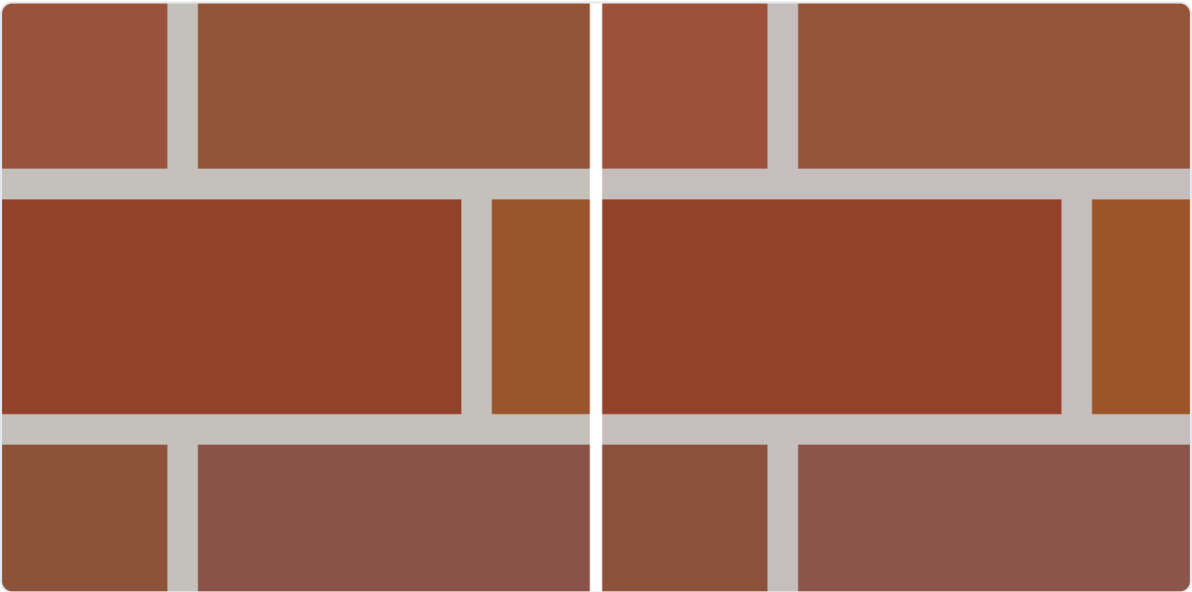
Clouds: original vs S3TC.

Issues with S3TC

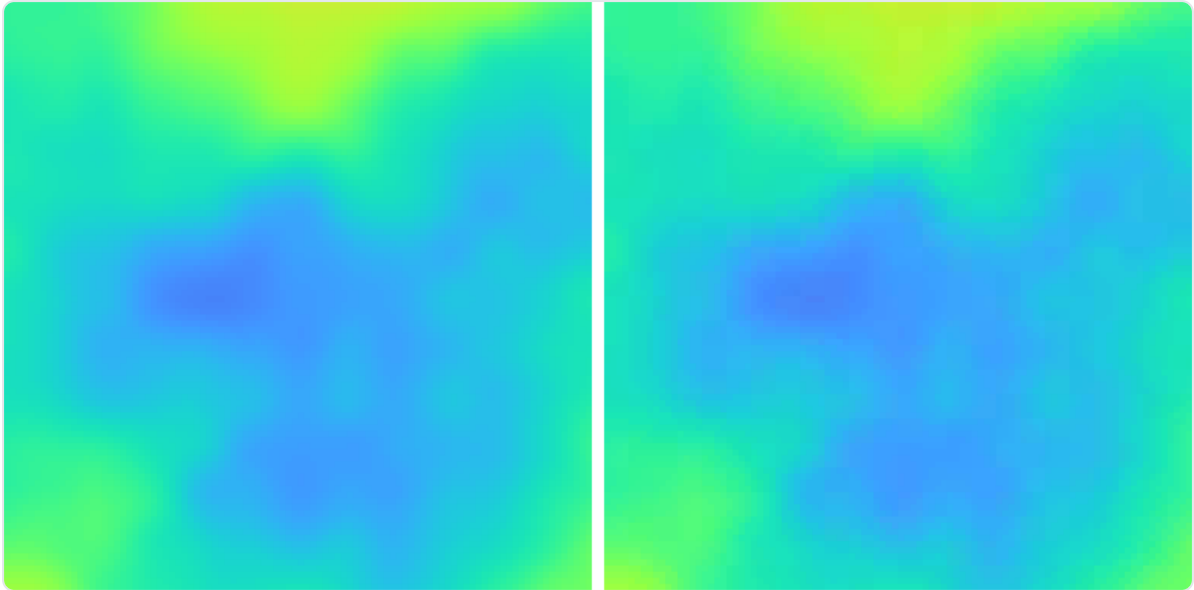
Isotropic colorings (i.e. large variances in the second and third PCA axes) cause problems for S3TC. The palette is four points on the line between two endpoints, so if the colors vary in two dimensions inside a block, the approximation is inaccurate. Additionally, due to the nature of S3TC, edges not aligned to the 4×4 grid are smeared or stepped, and every block gets the same 64 bits whether it needs them or not, leading to more inefficiencies. Also notable is that S3TC inaccuracies are made more visible by steps at every block boundary (very noticeable on the smooth gradient), making true perceived representation quality probably worse than the PSNR rating would suggest.



Gradient: block banding.



Bricks: blocky shading.



Clouds: faint blocking.

Why does green get an extra bit?

Green carries most of humans' perceived luminance (Rec. 709 weights ≈ 0.21 R, 0.72 G, 0.07 B), so a green error is $\sim 3\times$ as visible as in red and $\sim 10\times$ as visible as in blue. The extra level on green removes the most visible banding per bit. Evolutionarily, this is probably because sunlight is mostly green and because green wavelength is in between red and blue.

Why divide-by-levels quantization is a good approximation

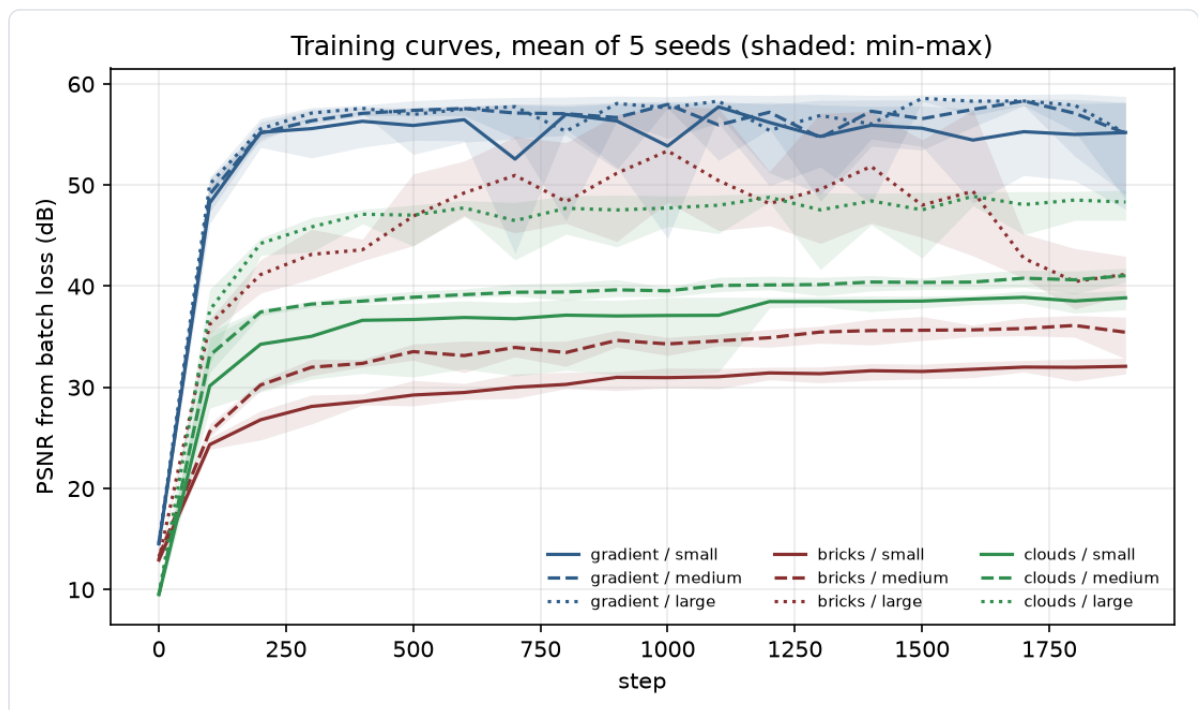
`q = round(c · levels)`, `c_hat = q / levels` returns an exact rational, while hardware reconstructs a code by bit replication into 8 bits ($\approx \text{round}(q \cdot 255/31) / 255$). Since $255/31$ is not an integer the two can differ, but by under a $1/255$ step, and both are monotonic with the same half-step rounding error, so the difference is not significant.

P6. Results and comparison

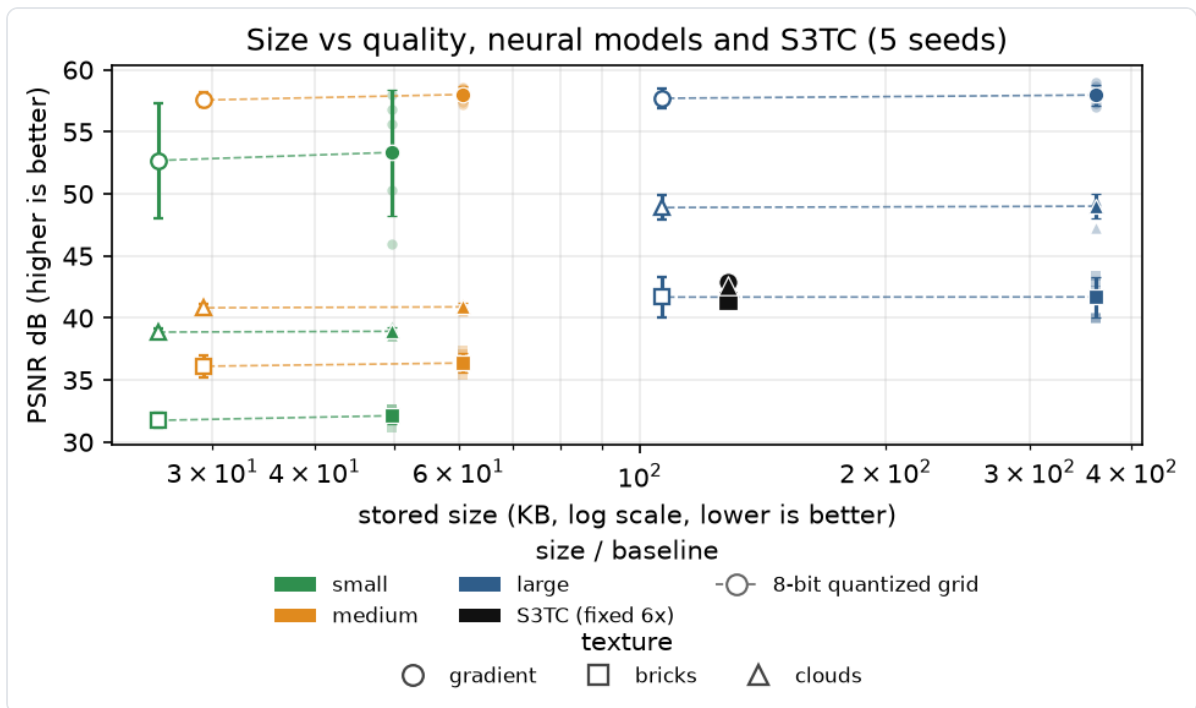
Five seeds per configuration; S3TC is the fixed $6\times$ point.

Texture	Model	PSNR (dB)	Size	Compression (raw/comp)
gradient	small	53.3	49.8 KB	15.4×

Texture	Model	PSNR (dB)	Size	Compression (raw/comp)
gradient	medium	58.0	60.8 KB	12.6×
gradient	large	58.0	361.3 KB	2.1×
gradient	S3TC	42.9	128.0 KB	6.0×
bricks	small	32.1	49.8 KB	15.4×
bricks	medium	36.4	60.8 KB	12.6×
bricks	large	41.7	361.3 KB	2.1×
bricks	S3TC	41.4	128.0 KB	6.0×
clouds	small	38.9	49.8 KB	15.4×
clouds	medium	40.9	60.8 KB	12.6×
clouds	large	49.0	361.3 KB	2.1×
clouds	S3TC	42.6	128.0 KB	6.0×



PSNR over training, all nine combinations (mean of five seeds, shaded band min-max).



Color: size or baseline, marker shape: texture. Filled = float32, open = 8-bit quantized grid

Best architecture per texture

- **Gradient: medium (58.0 dB).** It matches large at a sixth of the size; a smooth ramp does not need the large grid's extra capacity. All sizes beat S3TC by ~10 dB or more.
- **Bricks: large (41.7 dB).** The sharp edges need the finest level and more features; small collapses to 32.1 dB because one 64-level grid cannot hold mortar edges.
- **Clouds: large (49.0 dB).** Only large has coarse and fine levels plus enough features for the multi-scale detail; medium is 40.9 dB.

Isolating effects: small→medium (extra coarse levels) gains +4.7 / +4.3 / +2.0 dB (gradient/bricks/clouds); medium→large (feat_dim 2→4 plus the 128 level) gains +0.0 / +5.3 / +8.1 dB.

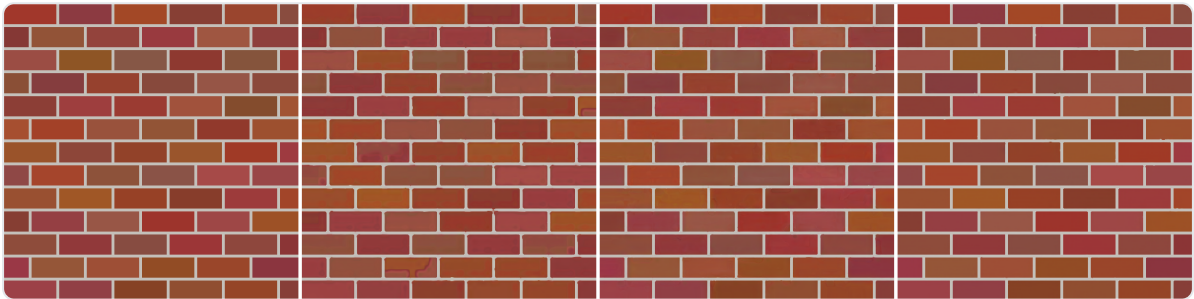
Did each texture's compression beat S3TC?

- **Gradient: yes.** All sizes beat S3TC, and small is both smaller (15.4× > 6×) and higher PSNR.
- **Bricks and clouds: no clean win.** Only large beats S3TC on PSNR, and it is larger (2.1× vs. 6×). Small and medium are smaller but lower PSNR. At a matched size S3TC is better here.

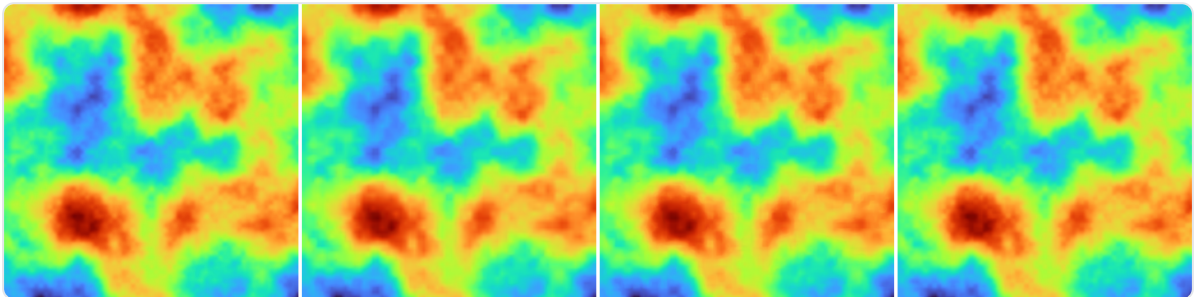
Reconstructions



Gradient: original, small, medium, large.



Bricks: original, small, medium, large.



Clouds: original, small, medium, large.

P7. Quantization

Post-training, each stored array is quantized separately over its own range: round to the nearest of 256 levels, store q , decode $\hat{x} = l_o + q \cdot \text{scale}$ (one byte per value plus two float32 range endpoints per array).

Texture	Model	PSNR float	PSNR 8-bit
gradient	small	53.3	52.4
gradient	medium	58.0	56.1
gradient	large	58.0	57.1
bricks	small	32.1	31.3

Texture	Model	PSNR float	PSNR 8-bit
bricks	medium	36.4	35.4
bricks	large	41.7	41.5
clouds	small	38.9	38.4
clouds	medium	40.9	40.3
clouds	large	49.0	48.1

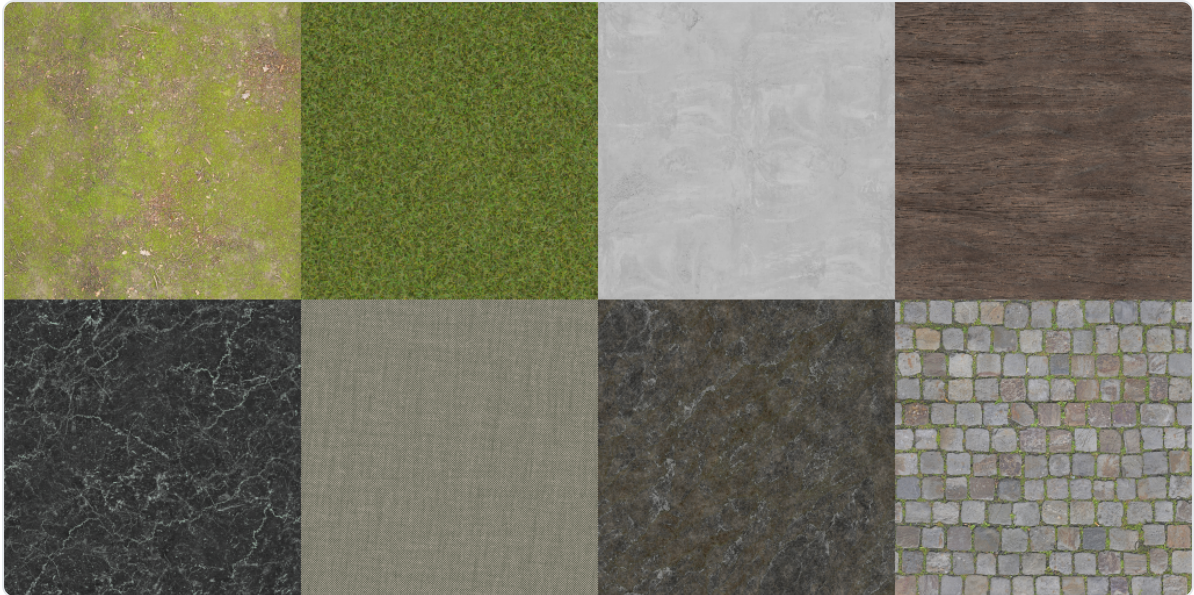
8-bit column quantizes both the grid and the MLP; quantizing the grid alone costs 0.01–0.65 dB.

Model	float32	8-bit grid	8-bit grid + MLP
small	49.8 KB (15.4×)	25.8 KB (29.8×)	12.5 KB (61.5×)
medium	60.8 KB (12.6×)	29.3 KB (26.2×)	15.3 KB (50.3×)
large	361.3 KB (2.1×)	106.3 KB (7.2×)	90.4 KB (8.5×)

Quality lost: 0.01–0.65 dB when only the grid is quantized (large under 0.3 dB); up to 1.9 dB on the small model when the decoder is quantized too (about 0.9 dB on medium/large). The grid features span a narrow range, so a bucket is a tiny fraction of the value scale and the error is a small uniform rounding, so the quantization just turns into noise. The small model is affected most because its MLP is a large share of its size and has a wider weight range.

P8. Textures sourced online

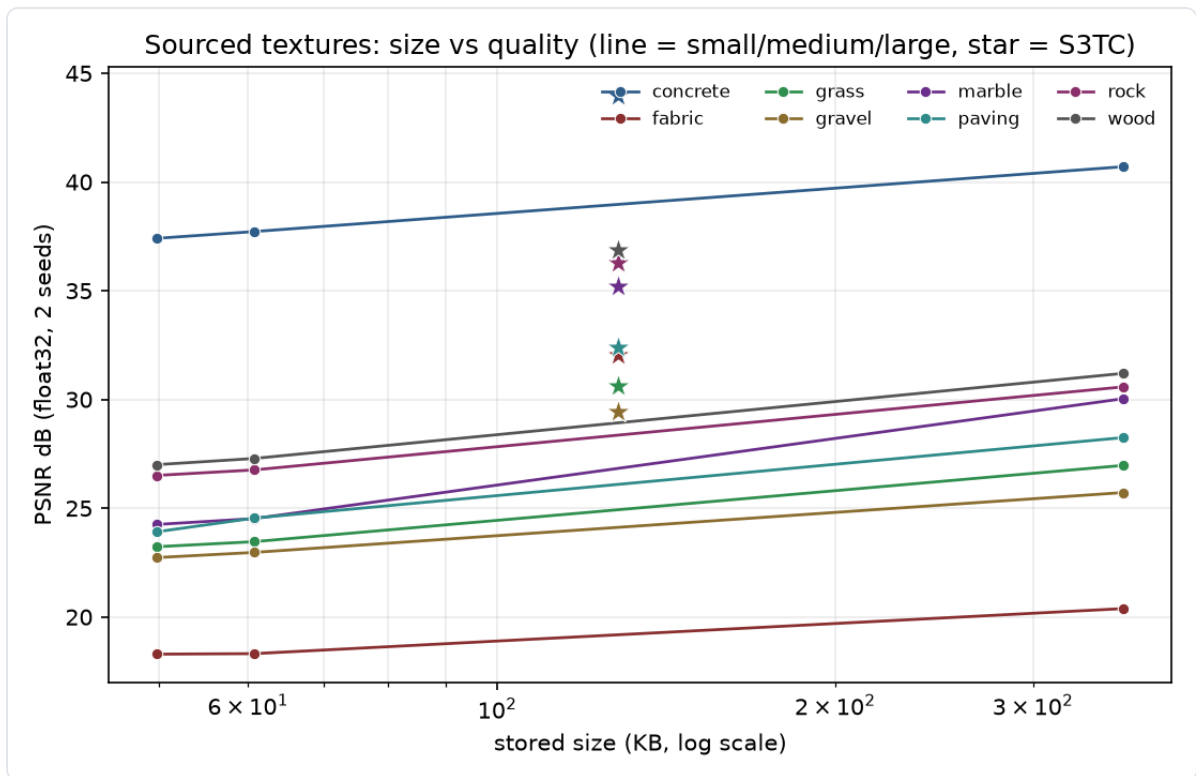
I found eight CC0 color maps from ambientCG spanning frequency content. Relative to the given textures, they have less-varying colors and more high-frequency patterns. I used the mean absolute Laplacian of the grayscale texture as a detail metric (higher = more high-frequency energy).



Top: gravel, grass, concrete, wood. Bottom: marble, fabric, rock, paving.

Texture	detail score	S3TC	small	medium	large
concrete	0.036	44.0	37.4	37.7	40.7
wood	0.112	36.9	27.0	27.3	31.2
rock	0.113	36.3	26.5	26.8	30.6
marble	0.124	35.2	24.3	24.5	30.0
paving	0.139	32.4	23.9	24.5	28.2
grass	0.138	30.6	23.2	23.5	27.0
gravel	0.178	29.4	22.7	23.0	25.7
fabric	0.477	32.1	18.3	18.3	20.4

Size per architecture: small 49.8 KB (15.4×), medium 60.8 KB (12.6×), large 361.3 KB (2.1×); S3TC 128 KB (6×).



Each line is small→medium→large for one texture; the star is S3TC.

On every sourced texture, the baseline S3TC beats the neural representation, including the much larger large model.

- **Fine repetitive weave (fabric, detail 0.477):** The weave sits near the finest grid resolution, and rather than dropping it the model invents smooth large-scale blobs that are not in the texture: a bilinear grid is a low-pass filter, and content above its cutoff turns into spurious low-frequency structure.
- **Sparse thin veins (marble, 0.124):** Small and medium blur the veins into wavy bands; large gains +5.5 dB because the 128 level and four features can trace them, but it is still behind S3TC.
- **Fine stochastic detail (gravel 0.178, grass 0.138, paving 0.139):** On mid-detail textures, S3TC still wins by 3.6–4.2 dB; the grid smooths the fine structure.
- **Directional and multi-scale (wood 0.112, rock 0.113):** S3TC wins by 5.7 dB; large recovers much of the structure that small/medium lose.
- **Smooth low-frequency (concrete, 0.036):** This texture is the closest to the starter gradient, but it is still 3.3 dB behind S3TC; the reconstruction loses the fine aggregate speckle. The feature grid + MLP implementation's success over S3TC seems to be specific to textures far below the grid's bandwidth, not a general improvement.

Conclusions

- S3TC is a strong baseline: a fixed 6× at 41–44 dB PSNR, but its four-colors-per-block palette limits color range and leaves visible blocking on smooth gradients.
- The feature grid + MLP representation dominates when the content is smooth and low-frequency and the color range is wide: on the starter textures it beats S3TC, and after 8-bit quantization the large model beats it on all three at a smaller stored size (90.4 KB vs 128 KB).
- It loses on natural textures. Across all eight sourced textures S3TC wins, even against the much larger large model, and the gap grows with high-frequency detail (3.3 dB on concrete up to 11.7 dB on fabric). It behaves as a low-pass representation: detail it cannot carry becomes blur, and content near or above the grid cutoff can fold into spurious low-frequency structure.
- Quantization is cheap. Quantizing the grid costs under 0.65 dB and shrinks the representation 2–4×; the small model reaches 61.5× at 52.4 dB on the gradient.
- The two methods are complementary: the learned model is better for smooth, wide-color-range content, while S3TC is better for fine detail. Closing the high-frequency gap needs higher grid resolutions and more levels, more feature channels, and maybe positional or hash features to improve the learnability of fine detail.
- In the future, I would like to experiment with larger feature grids to learn higher-frequency details. I would also like to experiment with more aggressive quantization because I haven't hit an amount of quantization where the neural representation breaks down and the PSNR is barely affected at the current levels of quantization.

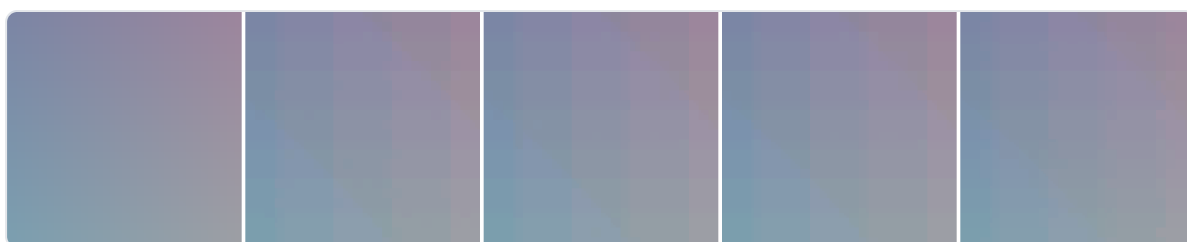
Appendix: seam-aware S3TC (beyond the required problems)

Side experiment: how much of S3TC's blocking is recoverable by changing only the endpoint choice at the same 8 bytes/block? The seam metric compares, for every adjacent texel pair straddling a block boundary, the mean absolute difference between the reconstruction's color step and the original's, so a perfect reconstruction scores 0 and only added discontinuities count. Baseline seam: gradient 0.00225, clouds 0.00690, bricks 0.00059. About half the gradient seam is RGB565 rounding alone (float endpoints drop it to 0.00119).

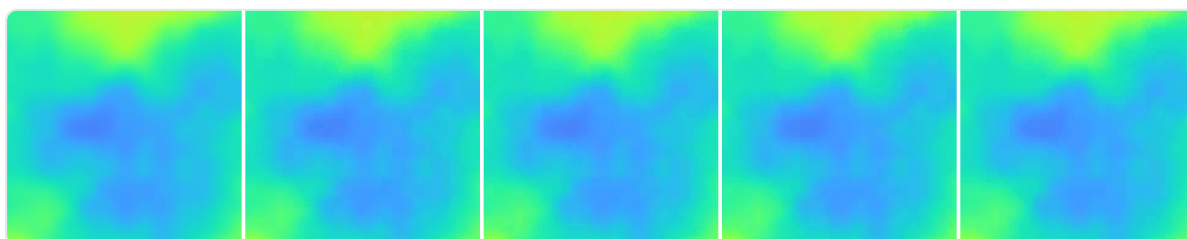
I tested with a few different methods of optimizing the baseline endpoint selections by creating different loss functions that took into account a block's neighbors and running gradient descent-like optimization on them.

- **Block error diffusion** feeds each block's edge error into its right/below neighbors before fitting them. Mild: gradient seam -8% , clouds -4% ; needs a light touch (alpha 0.1).
- **Boundary-weighted refit** pulls edge texels toward `neighbor_decoded + original_step`. Best seam on the gradient (-13%), roughly neutral PSNR.
- **Seam-energy ICM** lets each block search candidate endpoints (its own, PCA, the baseline fit, and blends toward neighbors) against a local error plus a seam penalty. Best overall: gradient $+1.34$ dB and -9% seam, clouds $+0.16$ dB and -14% seam.

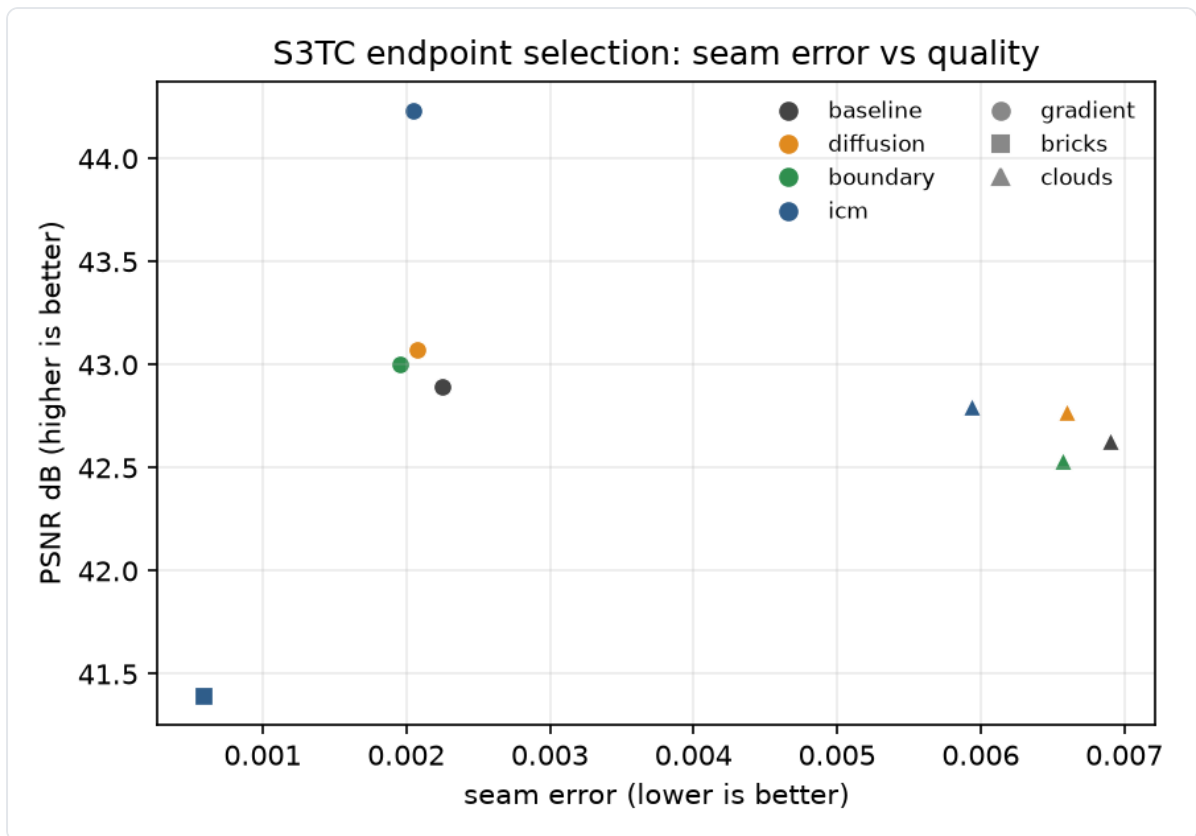
All keep the fixed $6\times$, and none remove the roughly half of the seam that is RGB565 rounding. Bricks is unchanged by all three (its blocks are flat, so there is no added step to fix), a useful sanity check.



Gradient, $4\times$ zoom. Left to right: original, baseline, diffusion, boundary, ICM; the banding gets progressively less pronounced.



Clouds, $4\times$ zoom, same order.



PSNR against seam error for the three methods and the baseline; up and to the left is better.